



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

–1 POINTS FOR EACH FIELD LEFT EMPTY OR FILLED WRONG

FIRST NAME: LAST NAME:

ID NUMBER: SECTION:

STUDENT SIGNATURE: ROOM NUMBER

MIDTERM EXAMINATION

COURSE NAME: COURSE NUMBER:

EXAMINATION DURATION:

ADDITIONAL MATERIALS ALLOWED TO BE USED:

SPECIAL INSTRUCTIONS

(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS. COMPLIANCE IS REQUIRED! DO NOT ARGUE WITH A PROCTOR, YOU GET "F" FOR MISBEHAVING + other possible complications.

(2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET, YOU MUST PUT ONLY YOUR ON EVERY ODD PAGE (do NOT write your name)

- Final answers must be written by **only blue or black, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

Do NOT fill	P.1	P.2	P.3	P.4	P.5	P.6	Total
max points	5	3	6	8	3	5	30
your score							

1 Logic

1. (5 points) Determine the truth value of each of these statements if the domain for all variables consists of all integers. Write **T** or **F** in the corresponding box next to the statement.

- $\forall x \exists y (x^2 < y)$
- $\exists x \forall y (x < y^2)$
- $\exists x \forall y (xy = y)$
- $\exists x \exists y (x^2 + y^2 = 6)$
- $\exists x \exists y (x + y = 4 \wedge x - y = 1)$

2. (3 points) Find DNF of $(x \oplus y) \wedge z$

2 Sets

3. (6 points) Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

- $A \cup B$

- $B \setminus A$

- any one-to-one function $f : B \rightarrow A$

- any onto function $g : A \rightarrow B$

3 Divisibility and Modular Arithmetic

4. (8 points) Solve the linear congruence

$$89 \cdot x \equiv 2 \pmod{232}$$

4 The Growth of functions

5. (3 points) Let $f(x) = x^4 + \ln(3x^8 + 7)$ and $g(x) = (x^2 + 4x - 1)^2$.

Determine if the statement is true or false. Write **T** or **F** in the corresponding box.

• $f(x) = O(g(x))$

• $x^2 = O(g(x))$

• $f(x) = \Theta(x^4)$

• $g(x) * f(x) = O(x^4)$

• $f(x) = \Theta(g(x))$

• $g(x) + f(x) = O(x^4)$

6. (5 points) Prove that $f(x) = x^4 + \ln(3x^8 + 7)$ is $O(x^4)$

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